NLP · RECOMMENDATION SYSTEMS · PERSONAL PROJECT

Content-Based Movie Recommendation Engine

Recommends similar movies by comparing their storylines, without needing any user history.

Built using scraped IMDb storylines · Kasthuri Sathish

The problem

With thousands of titles, people struggle to find a movie similar to one they liked. New users have no viewing history, so recommendations based on past behavior don't work for them.

What I built

A recommendation engine that scrapes movie storylines and metadata, measures how similar the stories are, and suggests related titles through a Streamlit search app.

How it works

Scrape IMDb storylines → Clean and prepare text → TF-IDF vectors → Cosine similarity → Top similar titles → Streamlit search app

Key product decisions

- Chose a content-based approach, so recommendations work for new users with no history.
- Used storylines as the main signal, because they describe what a movie is actually about.
- Showed similarity scores, so users can see how close each recommendation is.
- Chose TF-IDF over heavier deep learning models for fast, lightweight similarity search with low computing cost.

Testing and learnings

- Tested with 20 popular film titles, checking that recommendations shared core themes, genres, and storylines.
- Removed stop words and punctuation so common words wouldn't inflate similarity scores.
- TF-IDF matches shared words, not meaning, so stories with different wording can be missed.
- Limitation: it can't learn personal taste the way user-behavior models can.

Tech stack

Python · NLP · TF-IDF · Cosine Similarity · Web Scraping · Streamlit

GitHub: github.com/kasthuri-tech/IMDB-Movie-Recommendation-System · Portfolio: kasthuri-tech.github.io